PAPER_1599_CARBON_STEEL_DENSITY_7850

Daniel T. Murphy

PAPER_1599 — Carbon Steel Density = ρ_steel = D_crit²·SO_5 + SO_5³ + SO_5·N_CH = 6760 + 1000 + 90 = 7850 kg/m³

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Engineering
Date: June 18, 2026
Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S568 (Engineering Unified Proof Set) lists **Carbon Steel Density** with the integer-primitive expression:

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```
ρ_steel = D_crit²·SO_5 + SO_5³ + SO_5·N_CH = 6760 + 1000 + 90 = 7850 kg/m³
```

``

Residual: **EXACT**.

UQFF Closed Identity

``

```
ρ_steel = D_crit²·SO_5 + SO_5³ + SO_5·N_CH = 6760 + 1000 + 90 = 7850 kg/m³
EXACT
```

``

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Y_S568
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "carbon_steel_density_7850"})`
